

Team 1: Maize and Blue Cupboard

Final Report

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### Background and Context

The Maize and Blue Cupboard (MBC) is a free food pantry operated through the University of Michigan's Office of Financial Aid. It is available to any student, staff, or faculty member with an active MCard and serves as one of the primary campus resources for students experiencing difficulty accessing food. Our team's project builds off of previous efforts by a 2021 ME450 capstone team that focused on the same service. The ME450 project team started with a wider interest in food insecurity and its causes. Through initial research and stakeholder identification, they scoped the project to MBC as the key campus resource addressing the issue. By engaging with the MBC, they were able to ground the project in a real organizational context with existing users and constraints, and they framed the problem primarily as one of usability, logistics, and communication, identifying issues such as unclear information about hours, eligibility, and available services. Based on this framing, they proposed the mobile application as a centralized platform to streamline information and make MBC more accessible. While this approach addressed some practical barriers, it treated the challenge as an interface and information problem, focusing on improving the artifact itself rather than the broader system of student behaviors, perceptions, and organizational constraints. As a result, deeper issues such as stigma, social dynamics, and long-term adoption were not addressed. In addition, the project had no plans moving forward after the semester ended, which resulted in us having a difficult time reaching out to the correct people to access the files. The project also did not have plans for updates and maintenance, which creates additional workload for the MBC if they want to use the app and keep it running. These observations shaped how we entered the project and what questions we chose to prioritize.

With the gaps we identified from the previous ME450 project, combined with insights from prior research on college food insecurity and design interventions, we guided our project with a different set of assumptions. First, the literature emphasized that food insecurity is a complex, systemic issue that requires an updated and context-specific understanding, which leads us to assume that the existing user data may be inaccurate due to the ME450 project being from five years ago. Second, multiple studies highlighted that stigma and emotional barriers, such as feelings of embarrassment or undeservingness, play a significant role in whether students access food resources. Combined with our personal experience, we hypothesized that negative stigma surrounding food insecurity is a major barrier to MBC usage. Lastly, the literature also emphasized the importance of iterative, sustainable design approaches rather than a one-off solution, which informed our assumption that a longevity guide is needed. This was further reinforced by the observation that the ME450 project did not continue beyond its initial development, indicating a gap in long-term usability and adoption.

These assumptions were validated through contact with Kelly O'Mara, program manager of MBC, and previous contact with the ME450 capstone team. During our initial interview, Kelly confirmed that despite the awareness of MBC improving, many students still choose not to use the food pantry because of perception-based barriers, noting beliefs such as that they are "not food insecure enough" to use the service and that someone else needs the resources more than them. She also emphasized on reframing food insecurity as food access, which aligns with our literature-informed focus on stigma and meaning. Additionally, Kelly stated that MBC lacks insight into non-users, which confirmed our assumption that an updated and broader user needs assessment is necessary. Lastly, she highlighted long-term sustainability of projects as a major challenge, as most student-led projects fail to persist

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beyond the semester and become an additional workload for MBC staff. This supports our assessment of the need for a reusable framework and transition plan.

Overall, the alignment between our literature-based assumptions and Kelly's insights confirmed that our project is well grounded in both research and real-world needs, which suggested that a more effective approach should look beyond a single artifact and engage more with the broader system that involves student perceptions, organizational constraints, and long-term usability.

### Project Goals

The overarching goal of this project was to help the Maize and Blue Cupboard develop a clearer understanding of why students who could benefit from the pantry are choosing not to use it, and to do so in a way that the organization can sustain and build on over time. Rather than creating another tool aimed at surface-level information barriers, we wanted to orient the project around understanding deeper factors that may shape whether students engage with campus food resources

This meant focusing the work on two different, but connected priorities. The first priority was to develop a more structured approach to understand the barriers students face in relation to using the MBC, with particular attention to any psychosocial or perception-based factors. This decision was made because the MBC had taken strides to better understand users' needs in recent years, implementing a "MBC Shopper Survey" every semester to learn more about how to improve the service. Their program manager, Kelly, however, articulated that a growing need was to understand why students haven't considered using the MBC, expressing a desire to refocus the issue towards *food accessibility* and not *food insecurity*. This difference encouraged us to pursue an updated needs assessment that can measure awareness or satisfaction among existing users, but also explore deeper barriers associated with the MBC.

Our other priority was sustainability, because Kelly was clear that one of the most persistent challenges that the MBC faces with student-led projects is that they do not continue beyond the semester mark. The deliverables either required ongoing maintenance that MBC staff could not absorb, or they were handed off without enough context for anyone to use them. This concern, reinforced by what we observed with the ME450 project's lack of a transition plan, led us to make longer-term usability a core project goal because we want to make sure that whatever we produced could be picked up and used by future student teams or MBC staff without requiring our involvement to explain or maintain it.

Because of this, our project goal was not to create another intervention for MBC. Instead, we approached artifact creation as a process of building a tool that the MBC can use to better understand and respond to the needs of students who are not currently engaging with the service. The primary artifact of this project is a survey instrument designed to be sent out campus-wide to capture the experiences, perceptions, and barriers of both users and non-users of the pantry. Along with the survey, we aimed to develop a longevity guide to document the reasoning behind the instrument, how to deploy it, and how to adapt it if MBC's needs change over time. Together, these artifacts were designed to give the MBC a way to collect meaningful data about food access on campus rather than a one-time snapshot that loses relevance after the semester. By focusing on the specific gaps Kelly identified, particularly the lack of insight into non-users, the role of stigma or perception in discouraging use, and the pattern of student

projects that do not persist, we aimed to produce work that supports MBC's ability to make informed decisions going forward without adding to their existing workload.

### **Design Science Approach**

This project follows a design science research (DSR) approach, which focuses on addressing real-world problems through the iterative design, development, and evaluation of artifacts. Rather than only analyzing the Maize and Blue Cupboard (MBC) as an existing system, our team aimed to design improved interventions grounded in both stakeholder needs and relevant theory. Design science emphasizes the creation of actionable solutions while maintaining rigor through literature integration, structured ideation, and reflective evaluation.

#### *Relevant Literature and Theoretical Grounding*

Our approach was informed by literature on food insecurity, stigma, and human-centered design, which highlights that access to resources is not solely an informational problem but also a social and psychological one. Meza et al. (2019) found that feelings of shame and perceived unworthiness prevented students from seeking food support even when they were aware of available resources, and El Zein et al. (2022) reported that many students avoided campus food pantries because they believed their need was not severe enough compared to others. Taylor et al. (2024) further demonstrated that the stigma of being food insecure and the stigma of accessing a food support service are distinct psychological experiences, reinforcing that interventions need to account for both. Broton and Goldrick-Rab (2016) argued that food assistance is more effective when it is integrated into everyday campus services rather than presented as a standalone program, which supported Kelly's emphasis on reframing food insecurity as food access. Together, these findings directly challenged the original framing of the MBC problem as primarily an awareness or usability issue.

In addition, our work was guided by design science principles outlined by Johannesson and Perjons (2014), particularly the emphasis on artifact creation, iterative refinement, and grounding design decisions in both theory and empirical insights. We also drew on the "Beyond the Artifact" perspective articulated by Hevner and Chatterjee (2010), which emphasizes that effective design extends beyond a single tool or interface and must account for broader systems, including user behavior, organizational constraints, and long-term sustainability. Bennett and Vokoun (2022) reinforced this through their framing of food insecurity as a wicked problem, arguing that designers should apply systems thinking and look at how different actors and resources interact across a food system rather than focusing on a single program or solution. This perspective was critical in shifting our focus from simply improving the existing app to rethinking the overall service experience and system surrounding MBC.

#### *Idea Generation and Down-Selection*

Following design science methodology, we engaged in a structured ideation process to explore a wide solution space before converging on a refined direction. Each team member initially generated three independent ideas, resulting in a diverse set of potential interventions. These ideas ranged from refining

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the existing mobile application through usability testing to integrating AI-powered features, such as personalized grocery assistants, to rethinking the problem through updated stakeholder research and journey mapping.

Building on these individual contributions, the team collaboratively expanded and synthesized these concepts into eight broader ideas. These included revisiting the needs assessment, conducting updated usability testing, analyzing comparable systems such as the Michigan Dining app, mapping user journeys, and exploring emotional and stigma-related barriers to engagement. This stage emphasized divergence and creativity, ensuring that we did not prematurely converge on a single solution and instead considered both technological and non-technological interventions.

From this expanded set, we narrowed our focus to three primary directions:

1. A **student-centered needs assessment** to validate current assumptions and identify updated barriers
2. **Journey mapping and persona development** to better understand lived experiences and emotional factors
3. A **critical theory-informed framework** to guide equitable and socially aware design decisions

This down-selection process reflects a core principle of design science: iteratively refining both the problem and the solution space based on evidence, rather than assuming a fixed direction from the outset.

### *Participatory and Human-Centered Focus*

A key component of our design science approach was a participatory, human-centered perspective. Rather than designing solutions based solely on assumptions or prior work, we emphasized the importance of engaging directly with stakeholders and centering their lived experiences. Perez et al. (2022) demonstrated that co-design activities with people who have experienced food insecurity can lead to meaningful changes in how participants communicate about and engage with food access challenges, showing that the process of involvement itself can be as valuable as the tools it produces. Johnson et al. (2025) similarly grounded their food insecurity communication project in a critical design framework that centered the perspectives of directly affected and historically excluded students throughout every stage of the design process. Following these examples, our approach included stakeholder interviews, surveys, and focus groups designed to engage both users and non-users of MBC.

This approach was particularly important given the sensitive nature of food access on campus. Literature and stakeholder insights revealed that stigma and emotional barriers play a significant role in whether students choose to seek support, meaning that purely technical solutions may not fully address the problem (Meza et al., 2019; El Zein et al., 2022). Henry (2017) found that food insecurity is often normalized among college students as an expected part of the experience, which can prevent students from recognizing their own need or feeling entitled to help. By incorporating more participatory methods, we aimed to ensure that our designs were grounded in real student experiences and addressed the social and psychological dimensions of the problem rather than treating it as an information or interface issue.

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### *Design Artifacts and System-Level Thinking*

Consistent with design science, our project focuses on the development of design artifacts. However, informed by the “Beyond the Artifact” perspective, we intentionally expanded our definition of an artifact beyond a single mobile application. Instead, our artifacts include:

- Updated needs assessments and stakeholder maps
- A framework for incorporating critical theory into future design decisions
- Recommendations for system-level improvements, including communication strategies and service design changes

This broader view allows our work to address not only the interface-level challenges identified in the original ME450 project, but also systemic issues such as stigma, adoption, and long-term sustainability.

### *Reflection on the Design Science Process*

Throughout this process, a key shift in our thinking was moving from a solution-first mindset to a problem-reframing approach. Initially, it may have seemed intuitive to refine or rebuild the existing app. However, through literature review, stakeholder insights, and iterative ideation, we recognized that the core issue was not simply awareness or usability, but a more complex combination of social, psychological, and systemic barriers.

This reflects the value of design science as a methodology: it enables teams to iteratively redefine both the problem and the artifact, ensuring that the final outcome is both theoretically grounded and practically relevant. Our approach ultimately positions the project not as the development of a single tool, but as the design of a more comprehensive and sustainable system for improving access to MBC resources.

## **Methods**

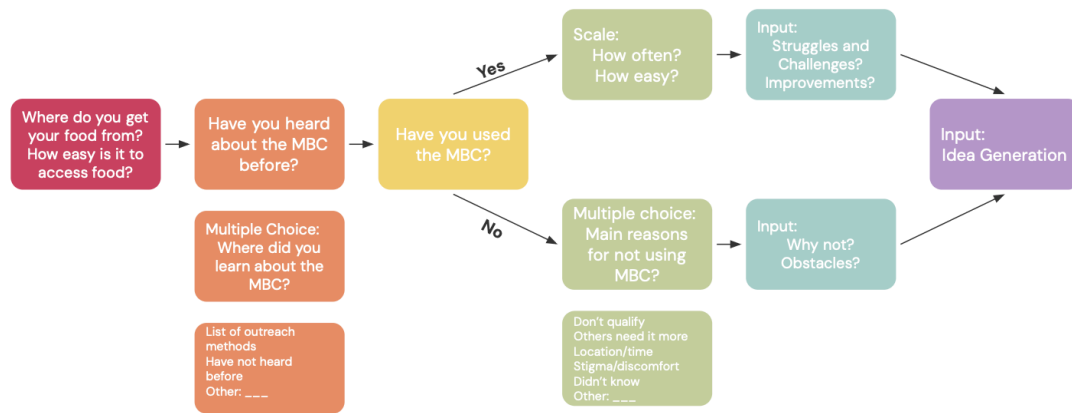
This section outlines the methods used to design, refine, and prepare the primary artifacts of this project within a design science framework. Our approach was guided by the need to create a solution that is both theoretically grounded and practically usable by the Maize and Blue Cupboard (MBC). Rather than relying on a single method, we combined literature-informed design, stakeholder engagement, and iterative internal evaluation to develop artifacts that address both the immediate need for updated user insights and the longer-term goal of sustainability. The methods described below focus on two key components: the development of the survey instrument as a needs assessment tool, and the creation of supporting resources to enable its effective deployment and continued use.

### *Survey Instrument Development*

The main artifact of this project is a campus-wide survey designed to understand how University of Michigan students experience food access and what issues may prevent them from using the Maize and

Blue Cupboard (MBC). We built the instrument grounded in the literature, aiming to better assess the way students may think about food access before/after learning about a specific resource, and keeping it short enough that students would be realistically more likely to complete it. A final copy of the survey questions is included in **Appendix A**.

We wrote survey items by pulling directly from constructs identified in our literature review and confirmed through our interview with Kelly. The survey flow is described in **Figure 1**, and opens with questions about everyday food behaviors, such as where students get most of their food and what makes accessing food difficult. The response options for the barriers question reflect categories that came up across multiple studies, including cost, time, transportation, kitchen access, and difficulty finding culturally or nutritionally appropriate food (El Zein et al., 2022; Broton & Goldrick-Rab, 2016; Henry, 2017). Before the survey mentions the MBC, it includes a block of Likert-scale items asking how students feel about food access in general. This ordering was one of the most intentional decisions we made. Taylor et al. (2024) showed that the stigma of being food insecure and the stigma of using a food pantry are two separate experiences. Items in this block ask whether students tend to keep food struggles to themselves, whether they see food access challenges as a normal part of college, and whether they think their peers talk openly about these issues. The non-disclosure items draw from Taylor et al.'s FISS scale, which found that the tendency to hide one's food insecurity status was the dimension most strongly linked to lower self-esteem and higher stress. Henry (2017) also influenced these survey items because they found college students often treat food insecurity as an expected trade-off of the college experience, which discourages them from seeking help even when they need it.



**Figure 1.** Survey Flow Logic

After the awareness questions, the survey branches based on whether the student has used the MBC. This structure lets us ask different questions of *users* and *non-users*, which responds directly to Kelly's observation that MBC has almost no information about why eligible students choose not to use the service. The 'non-user' path separates two types of stigma. Items about worrying about being seen or what pantry use would signal about one's financial situation capture anticipated stigma, the fear of social consequences if others found out. Items about feeling like one's situation is not serious enough capture self-stigma rooted in perceived deservingness, which El Zein et al. (2022) and Meza et al. (2019) both identified as among the most common reasons students avoid food pantries. This distinction matters

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practically because each type of stigma points toward a different design response. Anticipated stigma suggests features like discreet access could help, while self-stigma suggests that messaging about who the service is for may be more effective (Broton & Goldrick-Rab, 2016). The non-user section also asks about perceptions of food quality, since both El Zein et al. and Kelly confirmed that misconceptions about what MBC offers remain a common barrier. The ‘user’ portion instead asks about frequency, eligibility perceptions, and the psychosocial experience of using the service. Items like whether users felt comfortable and welcomed and whether they minded others knowing they used the MBC allow for comparison against the non-user stigma items.

The Likert-scale questions used a four-point scale (Strongly Disagree, Somewhat Disagree, Somewhat Agree, Strongly Agree) with the neutral midpoint removed. This was a deliberate choice. Chyung et al. (2017) found that respondents frequently use a neutral option as a way to avoid engaging with a question, especially when the topic is unfamiliar or uncomfortable. Combrinck (2024) similarly argued that removing the midpoint produces more precise data because it asks respondents to lean in one direction rather than default to the middle. Because our items ask about stigma, social discomfort, and deservingness, topics where avoidance and social desirability bias are well documented (Taylor et al., 2024; Meza et al., 2019), we wanted to reduce the likelihood that respondents would use a neutral option to sidestep questions they found sensitive. The four-point scale still offers gradation through the "somewhat" options while pushing toward more honest directional responses.

**Table 1.** Initial Requirements for Survey Instrument

| # | Requirement                                                                                    | Specification                                                                                                                                                                                                  |
|---|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | The survey should generate actionable information for MBC                                      | Includes structured questions on barriers, perceptions, and eligibility awareness for both users and non-users, with open-ended questions for additional context. Stakeholder goal is >150 survey respondents. |
| 2 | The survey should reach students beyond those already using the service                        | Distributed through general campus channels, framed around food access broadly, and designed to take 5 to 7 minutes to reduce drop-off                                                                         |
| 3 | The survey should surface how students think and feel about food access, not just what they do | Likert items ask about comfort, visibility concerns, perceived deservingness, and willingness to talk about food challenges                                                                                    |
| 4 | The survey experience should feel respectful and low-risk for respondents                      | Fully anonymous, uses "food access" language throughout, and avoids labels like "food insecure" that could feel stigmatizing                                                                                   |
| 5 | The survey should be easy for someone outside the team to understand and reuse                 | Built on a free platform (Qualtrics through U-M) and accompanied by a longevity guide with deployment instructions and design rationale                                                                        |



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An initial draft of the requirements and specifications, as determined by our literature review and stakeholder interviews, for the survey instrument is included above in **Table 1**. The initial survey was designed to take five to seven minutes. Research on survey completion in higher education settings shows that shorter instruments reduce dropout and improve participation, especially in online contexts where attention is competing with other demands (Holtom et al., 2022; Fosnacht et al., 2017). This was also a practical requirement given our distribution channels, where students encounter the survey alongside many other requests for their time. The survey collects no personally identifiable information, consistent with MBC policy and our agreement with Kelly to protect respondent privacy. The survey introduction also focuses on food access over food insecurity, based on both Kelly's guidance and the literature's emphasis on reducing stigma through more intentional language choices (Broton & Goldrick-Rab, 2016; Taylor et al., 2024).

Throughout the initial survey development, the team reviewed each item through multiple rounds of discussion to ensure the questions were being interpreted consistently and that each one was clearly connected to the problem we were trying to understand. After generating the initial items, we each took the survey individually and then met to walk through each question together, talking through how we read each item, where wording felt unclear or leading, and whether the overall flow made sense. This internal review process helped us catch issues that would not have been visible from the perspective of any single team member and ensured that the final instrument reflected a shared understanding of what we were measuring and why.

### *Longevity Guide Development*

To guide the development of our survey, we utilized the Design for Longevity Framework by Carlsson et al. This framework asserts that the optimal lifetime depends on using 6 main design strategies to balance what the article names as contextual factors: the user, the business, and resource efficiency. While the article applies this framework to assist designers in defining the best lifespan of their artefact, we believe that the overall framework design considerations would help create an artefact with the context of the MBC's needs. The framework also adapts Baker et al.'s (2019) six design strategies: 1) Design for Attachment and Trust, 2) Design for Durability, 3) Design for Standardization and Compatibility, 4) Design for Ease of Maintenance and Repair, 5) Design for Adaptability and Upgradability, and 6) Design for Dis- and Reassembly. These strategies were adapted for an online survey used to help assess the usefulness and reliability of our artefact. For example, in relation to strategy 5, we reflected and assessed whether the MBC team would be able to edit and change the survey as needed if the problem context changes.

We saw the MBC team that ran the survey as the users and aimed to design the survey to accommodate their needs, preferences, and perspectives to reduce the chances that the survey would become prematurely obsolete. We saw the entire MBC system as the business, considering how it would adapt around the implementation of the survey and how the survey can assist with the broader goals. Finally, we valued resource efficiency to be sustainable and reduce waste. We initially wanted to create an internal Longevity Guide for the MBC and future collaborators to help support future MBC projects. In the guide, we planned to describe our process and provide recommendations to future student teams. However, after speaking with Kelley, she described how she would prefer a plan to implement the survey and a write-up to improve engagement with the survey because previous surveys did not meet her target

respondents. As a result, we pivoted that deliverable from a longevity guide to a dissemination guide to meet her needs. She further stated that the MBC was better equipped to do future potential statistical analysis.

While the survey has not yet been deployed, the structure of the instrument allows us to anticipate the types of insights it is designed to generate. Based on both the literature and stakeholder input, we expect that barriers to using the Maize and Blue Cupboard will extend beyond awareness and logistics, with psychosocial factors such as perceived deservingness, discomfort with visibility, and normalization of food access challenges emerging as dominant themes among non-users. We also anticipate meaningful differences between users and non-users, particularly in how stigma is experienced and internalized. For example, non-users may report higher levels of anticipated stigma and self-stigma, while users may reflect more positively on the experience once barriers to initial engagement are overcome. In addition, the survey is designed to surface misconceptions about eligibility and food quality, which are expected to persist even among students who are generally aware of the service. These expected findings are not treated as conclusions, but rather as hypotheses embedded into the artifact itself, reinforcing the role of the survey as a tool for generating actionable, evidence-based insights once deployed.

### **Outcomes**

Within a design science framework, evaluation is critical to determining whether an artifact is both useful and usable within its intended context. Because the survey has not yet been deployed at scale, our evaluation focused on internal validation and planned real-world performance metrics. Internally, the survey instrument was iteratively reviewed and tested by all team members to ensure clarity, consistency in interpretation, and alignment with the constructs identified in the literature. This process helped identify ambiguous phrasing, improve question flow, and confirm that each item contributed meaningfully to the overall objective. Externally, the effectiveness of the artifact will be evaluated through its ability to generate actionable insights for the Maize and Blue Cupboard, measured through response volume, diversity of respondents, and the clarity of patterns in the collected data. The target of at least 150 responses serves as a baseline indicator of sufficient engagement, while the inclusion of both structured and open-ended responses enables both quantitative and qualitative analysis. Ultimately, the success of the artifact will be determined by whether it can be repeatedly deployed and used by MBC to inform decision-making without requiring additional design effort.

### *Dissemination Plan*

To ensure that the developed survey instrument generates meaningful and actionable insights, we designed a structured dissemination plan that enables broad, repeatable, and low-effort distribution across the University of Michigan student population. This plan was created in direct response to stakeholder needs, particularly the Maize and Blue Cupboard's (MBC) lack of access to large-scale, representative data, especially from non-users.

The dissemination approach leverages existing university communication infrastructure to maximize reach while minimizing the need for new systems. Distribution is structured across three primary channels:

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### 1. University-Level Channels

Institutional platforms such as Central Student Government (CSG) newsletters, College of Engineering (CoE) newsletters, Women in Science and Engineering (WISE) communications, and Canvas dashboard announcements are used to reach a large portion of the student body. These channels require structured submission processes (e.g., communication request forms, 7 to 10 day lead times for Canvas announcements) and allow for scalable distribution with minimal ongoing effort.

### 2. Departmental and Organizational Outreach

Targeted emails to academic departments (e.g., via department listservs such as me-aso@umich.edu) and student organizations enable engagement with specific student populations. This includes leveraging MCommunity to identify and reach demographic or interest-based groups, improving representation across the dataset

### 3. Peer-to-Peer and Passive Distribution

Flyers with QR codes and informal sharing mechanisms (e.g., word-of-mouth, group chats) support continued visibility. This approach aligns with stakeholder insight that MBC usage and awareness often spread through peer networks rather than formal channels, making this an important complementary pathway.

This multi-channel approach ensures both breadth and diversity of responses, which is critical for capturing the experiences of underrepresented or non-engaged student groups.

The dissemination plan is designed to be easily executable by MBC or future project teams.

- Timing: Deployment is recommended during the Fall 2026 semester (September to December).
- Materials: A complete dissemination package is provided, including:
  - A flyer with a QR code linking to the survey (**Appendix B**)
  - A motivational one-pager explaining the purpose and importance of the survey (**Appendix C**)
  - Pre-written email templates for outreach (**Appendix D**)
  - A curated list of newsletters, departments, and student organizations (**Appendix D**)
- Execution: Survey distribution is initiated via newsletter submissions and outreach emails, with targeted follow-ups if response rates are low.
- Monitoring and Closure: Responses are tracked through the survey platform, with a target of  $\geq 150$  responses. If participation plateaus, additional outreach efforts are triggered. The survey is closed once sufficient data has been collected.

A central goal of this dissemination plan is longer-term usability. To support this, all materials and processes are designed for easy transfer and reuse. MBC retains full ownership of the survey and dissemination assets, allowing them to deploy the survey independently in future semesters. By combining standardized templates, clear execution steps, and reliance on existing communication

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channels, this plan reduces dependency on individual project teams and supports continued data collection over time.

The current list of newsletters, departments, and student organizations should be considered a non-exhaustive starting point rather than a finalized resource. It was developed based on readily accessible and known communication channels, but does not fully capture the breadth of outreach pathways across campus, particularly among smaller or underrepresented student groups. Additionally, many of these channels are dynamic; submission processes, contact points, and levels of engagement can change over time.

As a result, this resource should be treated as a living document that requires ongoing iteration. Future teams or MBC staff should continue expanding the list through platforms such as MCommunity, updating contact information, and refining outreach strategies based on which channels generate the most responses. This continued development is critical to improving both the reach and representativeness of the collected data over time.

### **Design Science Contribution**

This project's contributions to design science are primarily through a practical lens, as we focused on the real-world usefulness of our process and ensuring that the final deliverables are useful, easy to adopt, and valuable for the Maize and Blue Cupboard. Here, the rigor of our project is the close alignment between the deliverables, design frameworks, the current literature on food access, and the desires of our main stakeholder. This project was guided by research on food access, particularly on college campuses, from the beginning, and throughout the development of the survey and the guide, we integrated multiple design frameworks and our individual experiences to turn the deliverables into valuable artefacts. When we got in contact with Kelly, the Maize and Blue manager, we used interviews to identify community stakeholder needs and translate them into design requirements and decisions. Even though our end deliverables did not go through a systematic design process, we are able to contribute to design science through reusable and transferable design knowledge that goes beyond the artefact and meets the needs of our stakeholders.

#### *Beyond the Artefact Design*

Beyond the artefact design pushes us, as designers, to consider how our artefact fits within the surrounding environment, instead of existing as a standalone entity. Our process of starting with the existing literature, interviewing Kelly, creating our survey, and iteratively refining our survey to meet Kelly's needs ensures that the survey fits into and adds value to the Maize and Blue Cupboard efforts to increase access to existing food resources. In the first interview, Kelly mentioned how there is an existing survey for people who have used the pantry in the past, but there isn't a survey for people who do not use it, which creates a gap in the MBC's understanding of why potential users may avoid the pantry. Our survey addresses this gap and supports their existing processes. We also focused on considering their cultural environment by working to structure and word the instrument to avoid stigmatizing and confusing language, reflecting the MBC's compassionate support for the overarching UofM community. Our survey

also helps their future, long-term goals by creating a way for them to systematically collect data to inform future interventions and improvements.

### *Participatory Design*

This project sees the Maize and Blue Cupboard as a participant and a co-creator. Instead of simply using the MBC to move forward with our own ideas, we took a participatory approach with our engagement with Kelly. We trusted what Kelly said in our interviews at face value and centered her experiences as a main contributor. As a team, we constantly kept her concerns in mind, especially the ones with previous student teams prematurely abandoning efforts without an official handoff. Constantly engaging with our stakeholder, both virtually and in person, allowed our project to stay on track. Through her engagement, we realized that some of the team's initial ideas were not relevant to or useful for the problems the MBC was facing. Also, she identified additional factors that the team did not consider in publishing the final survey.

Overall, our development of the survey and dissemination plan can serve as an example of what it means to utilize Participatory Design and Beyond the Artefact design to implement stakeholder feedback. This is especially useful for community organizations that want meaningful engagement despite being limited in manpower.

Beyond its immediate application to the Maize and Blue Cupboard, this project contributes more broadly to design science by demonstrating how needs assessment itself can function as a primary artifact in complex, socially driven problem spaces. In contexts where barriers are rooted in perception, stigma, and behavioral dynamics, traditional solution-focused artifacts may fail to address the underlying system. This work illustrates an alternative approach, where the design of a structured, theory-informed data collection tool becomes the intervention that enables future solutions. Additionally, the integration of dissemination planning as part of the artifact highlights the importance of considering not only what is designed, but how it is activated within a real-world system. Together, these contributions offer a transferable framework for designing stakeholder-informed, low-maintenance artifacts that support ongoing learning and decision-making in resource-constrained organizational settings.

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## Appendix A. Student Experiences with Food Access on Campus Survey

### *Introduction*

We are a student design team working to better understand how students at the University of Michigan experience food access on and around campus. All members of the University of Michigan community are welcome to participate. This survey is anonymous, takes about 5-7 minutes, and your responses will directly inform efforts to improve food-related resources available to students, such as the Maize and Blue Cupboard.

### *Section 1: Food Access on Campus*

Q1. Where do you get the largest percentage of your day-to-day food from? (Select all that apply)

- Grocery Store
- Dining Hall
- Restaurants or Takeout
- Convenience or Corner Store
- Family or Friends
- Food Bank or Pantry (e.g., Food Gatherers, Maize and Blue Cupboard)
- Other (open text)

Q2. Please indicate how much you agree or disagree with the following statements about campus food access. (Strongly Disagree / Somewhat Disagree / Somewhat Agree / Strongly Agree)

- If I'm having a hard time getting my day-to-day food, I don't really tell anyone
- Struggling with food access is a normal part of the college experience
- I think most students who have trouble getting food don't talk about it
- I feel comfortable talking to friends or classmates about challenges I have affording or accessing food
- It is easy to find food that meets my dietary (e.g., cultural, nutritional) needs near me
- I know about resources on campus that can help me with food access

Q3. Have you ever experienced difficulty accessing food while at U-M?

- Yes
- No
- Unsure

Q4. Which factors have made accessing food difficult for you? (Select all that apply) *Displayed only if Q3 = Yes or Unsure*

- Cost (not enough money)
- Limited time
- Transportation challenges
- Don't know how to cook with what's available



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- Lack of kitchen/storage access
- It is hard to find the kind of food I want (nutritional, cultural, etc.)
- Other (open text)

Q5. Before this survey, have you heard of campus food resources like the Maize and Blue Cupboard?

- Yes
- No
- Unsure
- Other (open text)

Q6. Where have you learned about the Maize and Blue Cupboard from? (Select all that apply) *Displayed only if Q5 = Yes*

- University email or website
- Social Media
- Friend or classmate
- Professor, advisor, or staff member
- Flyer or tabling event
- Orientation or welcome week
- Other (open text)

Q7. Before this survey, have you used the Maize and Blue Cupboard? (Select all that apply) *Displayed only if Q5 = Yes*

- Yes, I have gone to the MBC store
- Yes, I have visited a pick-up site
- No
- Unsure
- Other (open text)
- Other (open text)

### *Section 2: Perceptions and Experiences*

This section branches into two paths based on respondent answers to Q5 and Q7. Non-users and students who are unaware of the MBC are directed to Path A. Students who have used the MBC are directed to Path B.

#### Path A: Students Who Have Not Used the MBC

*Displayed if Q5 = No, Unsure, or Other, or if Q7 = No, Unsure, or Other*

Prompt: For those that have not used the MBC, please share your existing knowledge about the Maize and Blue Cupboard below.

Q8. Who do you think is allowed to use the Maize and Blue Cupboard? (Select all that apply)

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- Any local (Ann Arbor) resident
- Faculty and Staff
- Graduate students (Master's and PhD students)
- Undergraduate students
- Only students with meal plans
- Only students with demonstrated financial need
- Anyone with an active MCard
- I'm not sure
- Other (open text)

Q9. What are the main reasons you have not used the Maize and Blue Cupboard? (Select all that apply)

- I don't think that I need it
- I didn't know it existed
- Logistics issues (scheduling, signing up online, accessing MBC)
- I don't have transportation
- I feel other students need it more than I do
- I would feel uncomfortable using it
- I don't think it offers the food I would use
- Other (open text)

Q10. Please indicate how much you agree or disagree with the following statements. (Strongly Disagree / Somewhat Disagree / Somewhat Agree / Strongly Agree)

- I would worry about being seen by someone I know at the MBC
- I feel like my situation isn't serious enough to justify using the MBC
- Using the MBC is as easy as using other campus resources
- I don't think that the MBC would have the food I need
- The MBC only carries canned goods and other non-perishables
- I don't think that there are high-quality items at the MBC
- I am concerned about what using the MBC would say about my financial situation to others

Q11. What, if anything, would make the Maize and Blue Cupboard more accessible to you? Or, what would make you more likely to use the Maize and Blue Cupboard? (Open-ended)

Q12. Are there any other experiences that you'd like to share about food access on campus? If so, please explain. (Open-ended)

### Path B: Students Who Have Used the MBC

*Displayed if Q7 = Yes, I have gone to the MBC store or Yes, I have visited a pick-up site*

Prompt: Based on your experiences using the MBC, please answer the following questions below.

Q13. Who do you think is allowed to use the Maize and Blue Cupboard? (Select all that apply)

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- Any local (Ann Arbor) resident
- Faculty and Staff
- Graduate students (Master's and PhD students)
- Undergraduate students
- Only students with meal plans
- Only students with demonstrated financial need
- Anyone with an MCard
- I'm not sure
- Other (open text)

Q14. How often have you used the Maize and Blue Cupboard?

- Once
- Weekly
- Bi-weekly
- Monthly
- Semesterly
- As needed
- I don't use the Maize and Blue Cupboard
- Other (open text)

Q15. Please indicate how much you agree or disagree with the following statements. (Strongly Disagree / Somewhat Disagree / Somewhat Agree / Strongly Agree)

- I felt comfortable and welcomed when using the MBC
- I was able to find a convenient time to access the MBC
- I didn't mind if people I know found out I was using the MBC
- I would recommend the MBC to a friend
- I was able to find what I was looking for at the MBC
- I feel comfortable talking about challenges with food access on campus

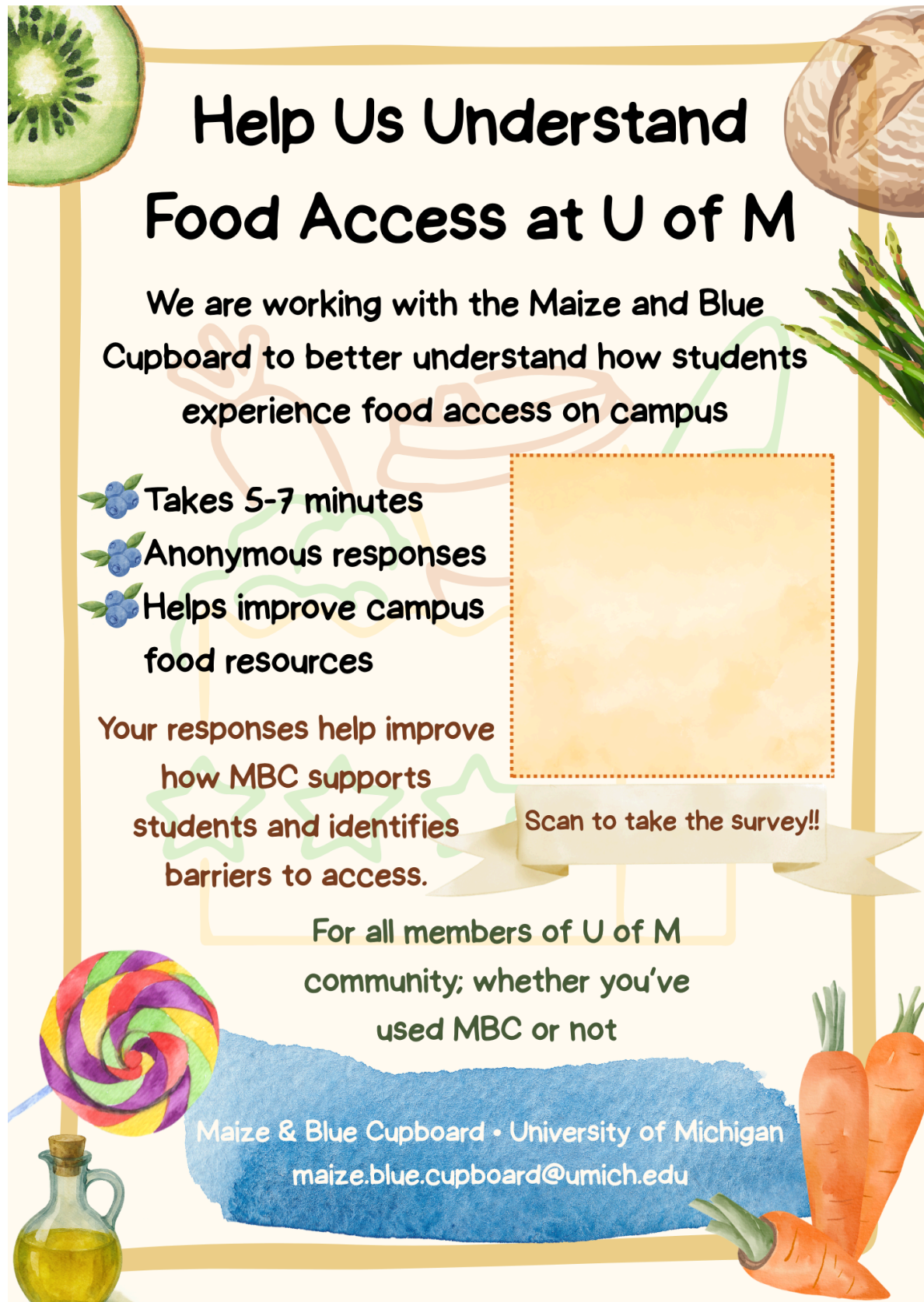
Q16. What suggestions do you have for improving the Maize and Blue Cupboard? (Open-ended)

Q17. Do you have any other significant experiences with the Maize and Blue Cupboard? If so, please share. (Open-ended)

### *Section 3: Closing*

Thank you for taking the time to share your thoughts with us about food access at the University of Michigan. Your responses will be kept anonymous and used to better inform future MBC initiatives.

Appendix B. Flyer for Survey Dissemination



# Help Us Understand Food Access at U of M

We are working with the Maize and Blue Cupboard to better understand how students experience food access on campus

-  Takes 5-7 minutes
-  Anonymous responses
-  Helps improve campus food resources

Your responses help improve how MBC supports students and identifies barriers to access.

Scan to take the survey!!

For all members of U of M community; whether you've used MBC or not

Maize & Blue Cupboard • University of Michigan  
[maize.blue.cupboard@umich.edu](mailto:maize.blue.cupboard@umich.edu)

## Appendix C. Motivational One Pager

# Help Understand Food Access at U of M

Past efforts to support the Maize and Blue Cupboard (MBC) focused on building tools to improve access to information, such as mobile applications. While well-intentioned, many of these solutions did not persist beyond the semester, leaving MBC without a sustainable way to continue using or building on the work. At the same time, both research and stakeholder conversations revealed a deeper issue: MBC lacks a clear understanding of why many students do not use available food access resources.

We found that these barriers are often not simply about awareness, but are shaped by stigma, perception, and uncertainty. Students may feel undeserving of support, unsure if they qualify, or uncomfortable engaging with food access services. Recognizing this, we shifted away from building another standalone tool and instead focused on understanding student behavior and experience more directly.

To address this gap, we developed a short, anonymous, campus-wide survey designed to capture both user and non-user perspectives. The survey focuses on awareness, perceptions, and behavioral barriers, and is intentionally structured to be reusable across semesters. Alongside the survey, we created a dissemination approach that leverages existing university communication channels (such as newsletters, departments, and student organizations) to reach a broad and diverse population.

This tool is intended to function as a recurring needs assessment for MBC. By deploying the survey once per semester and targeting a response size of at least 150 participants, MBC can generate meaningful insights into student needs and experiences. These insights can then be used to inform programming, improve outreach strategies, and better align services with the realities of the student population.

The significance of this work lies in shifting from one-time solutions to a sustainable system for ongoing learning. Rather than relying on assumptions or outdated data, this approach enables continuous, evidence-based decision-making. It also directly addresses a key challenge identified by MBC: ensuring that student-led work remains useful and actionable beyond the duration of a single project.

Moving forward, the impact of this effort depends on continued dissemination and engagement. Future teams, collaborators, and stakeholders can support this work by sharing the survey through their networks, expanding outreach to underrepresented groups, and refining messaging to reduce stigma and improve participation. With consistent use and thoughtful iteration, this system can provide lasting value to MBC and the broader student community.

## Appendix D. Dissemination Resources, Templates and Strategies

This document is intended to serve as a living resource. Contact points, submission processes, and engagement levels may change over time. Because the team that developed this resource in April 2026 was primarily composed of College of Engineering (CoE) students, the current dissemination strategies are biased toward CoE channels. Future teams or MBC staff should update and expand this list to improve reach across a broader set of campus communities.

### UNIVERSITY NEWSLETTERS

These channels provide broad visibility across the student body and should be prioritized for initial survey deployment.

#### Central Student Government (CSG) Newsletter

1. Submission Process: Complete the [CSG Communications Request Form](#)
2. Recommended Lead Time: 1–2 weeks
3. Required Materials:
  - a. Brief description of initiative
  - b. Finalized graphic (flyer with QR code)
  - c. Survey link

#### College of Engineering (CoE) Newsletter

1. Submission Process: Submit through the [OSA Student Announcements & Events Newsletter Submissions](#)
2. Required Fields:
  - a. Announcement Title: Help Us Understand Food Access at U-M (5-Min Survey)
  - b. Announcement Text: We are a student team working with the Maize and Blue Cupboard (MBC) to understand better how students experience food access on campus. This short, anonymous survey takes 5 to 7 minutes and will help improve campus food resources.
  - c. Link URL: [Insert Survey Link]

#### Women in Science and Engineering Newsletter

1. Submission Process: Submit via the [WISE Newsletter Announcement Request](#)
2. Notes:
  - a. Useful for reaching underrepresented groups in engineering
  - b. Consider tailoring messaging slightly for audience relevance

### DEPARTMENT-SPECIFIC NEWSLETTERS

These channels allow for more targeted distribution and often result in higher response rates.

#### Department Listservs

1. Contact Method: Email department administrators
2. Format: [department]-[me-aso@umich.edu](mailto:me-aso@umich.edu)
  - a. Examples: [me-aso@umich.edu](mailto:me-aso@umich.edu), [aerospace-aso@umich.edu](mailto:aerospace-aso@umich.edu)
3. Recommended Strategy:

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- a. Start with large departments (Engineering, LSA, Ross)
- b. Expand to smaller or identity-based groups for better representation

### **MCommunity Targeting**

1. Use MCommunity to identify:
  - a. Student organizations
  - b. Identity-based groups
  - c. Academic programs
2. Purpose:
  - a. Improve the diversity and representativeness of responses
  - b. Reach students less likely to engage with general newsletters

### **EMAIL OUTREACH TEMPLATE**

Use the following template for departments, organizations, and listserv requests.

**Subject:** Student Survey on Food Access at U-M

Hi [Department/Name],

We are a student team working with the Maize and Blue Cupboard (MBC) to understand better how students experience food access on campus.

We would greatly appreciate the opportunity to share a short (5 to 7 minutes, anonymous) survey with your students through your newsletter or listserv. The results will directly inform efforts to improve campus food resources and accessibility.

**Survey link:** [Insert Link]

**Flyer:** Attached

Thank you for your support,  
[Your Name / Team Name]

### **CANVAS DASHBOARD CAROUSEL**

1. Submission Process: Submit through [ITS-Canvas Tout Requests](#)
2. Lead Time: Minimum 7 to 10 business days
3. Requirements:
  - a. Image (rectangular format - cannot use flyer)
  - b. Short caption or description
  - c. Survey link
4. Notes:
  - a. High visibility but limited space → prioritize clear, concise messaging
  - b. Works best when paired with other outreach channels

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### **PASSIVE AND PEER-TO-PEER DISTRIBUTION**

These strategies complement formal channels and help reach students through informal networks.

#### **Flyers with QR Codes**

Can use the provided flyer with the QR code linked to the survey embedded in it

1. Place in:
  - a. Academic buildings
  - b. Libraries
  - c. Dining halls

#### **Informal Sharing**

1. Group chats
  - a. Lots of public GroupMe channels
  - b. Considerations in usage to avoid unnecessarily spamming
2. Student org Slack/Discord
3. Word-of-mouth
4. Rationale:
  - a. MBC awareness often spreads through peer networks
  - b. Helps reach students who may not engage with institutional communication

### **DEPLOYMENT STRATEGY**

To maximize response rate and diversity:

1. Launch Phase (Week 1–2):
  - a. Submit to university newsletters
  - b. Submit Canvas request
  - c. Begin department outreach
2. Expansion Phase (Week 2–4):
  - a. Target additional departments and student orgs
  - b. Increase flyer distribution
  - c. Share through peer networks
3. Follow-Up Phase:
  - a. Monitor response count
  - b. Re-send to high-impact channels if needed
  - c. Expand outreach to underrepresented groups

Target:  $\geq 150$  responses

### **NOTES FOR FUTURE USE**

1. This resource list is non-exhaustive and should be expanded over time
2. Track which channels generate the most responses
3. Update contact information regularly
4. Adapt messaging based on engagement and feedback